# Kulachart Parnduangkeaw

Data Science & Machine Learning · Computer Engineering, SWU

Pathum Thani, Thailand · +66 80 091 2939 · kulachart@kulac.dev · kulac.dev ·
linkedin.com/in/kulachart-parnduangkeaw-247b47395 · github.com/KhaNomThai



*Key results* **0.89 AUC-ROC** skin-cancer screening, 3,619 unseen images / **0.84 recall** peak-demand days vs 0.73 baseline / **Top 30** KBTG Hackathon 2026

## *Summary*

Computer Engineering student at SWU (expected 2027) and Data Science Intern at Botnoi Group (2026). Builds ML end to end, from leakage-safe validation and threshold tuning to models that run in the browser. Seeking a Data Science / AI internship.

## *Experience*

**Data Science Intern** — Botnoi Group · May – Jul 2026

- Built product demand forecasting for SME clients: data cleaning, feature engineering, model training and evaluation.
- Set up the MLOps pipeline to train and deploy the models, and built dashboards for the client companies.

## *Projects*

**Derma AI: Skin Cancer Screening** — CPE310 Healthcare AI · Solo project · 2026

- Fine-tuned ImageNet-pretrained ResNet50, EfficientNet-B3 and DenseNet121 on ISIC 2019 (25,331 images, 8 classes) with class-weighted loss for the imbalanced classes.
- Split by lesion (StratifiedGroupKFold) so no test lesion is seen in training; tuned each model's cancer threshold on validation to target 90% sensitivity.
- Best model: **0.89 AUC-ROC** and 88.7% top-2 accuracy on 3,619 unseen test images; exported to ONNX so it runs in the browser and images never leave the device.

Python · PyTorch · scikit-learn · Pandas · Deep Learning · Computer Vision · ONNX Runtime · React

**Peak Electricity Day Alert** — CPE312 Data Science capstone · 2026

- Forecast a week ahead which days hit peak demand from 2015–2020 Panama load and weather data, with leakage-safe daily features and a time-based train/test split.
- XGBoost caught **0.84 of peak days vs 0.73** for a same-day-last-week baseline, at higher precision (0.78 vs 0.73); also compared Logistic Regression.

Python · XGBoost · scikit-learn · Pandas · Matplotlib · Machine Learning

**Smart Flood AI** — DOH Hackathon 2026 · Team of 3 · Web & AI dev · 2026

- Zero-shot Grounding DINO finds cars and wheels in highway CCTV to **estimate water depth** from how far the wheels are submerged, with no labeled training data.
- Fed a FastAPI + Next.js command center with flood-aware routing and LINE OA alerts.

Python · PyTorch · OpenCV · Grounding DINO · FastAPI · Next.js · Firebase · Computer Vision

**KaTang Cash-flow Co-pilot** — KBTG Hackathon 2026 · Team of 3 · Data science · 2026

- Built the gradient-boosting spending forecast and Monte Carlo prototype (simulated data) behind a safe daily spending limit for variable-income first jobbers; **pitched in the top-30 round**.

Python · scikit-learn · Pandas · Machine Learning · Next.js · TypeScript

**Employment Data Gap Analysis** — Research project · 2025

- Random Forest on 100,000 placement records vs a survey of 80 CPE students: students weighted internships and portfolio above the model (**31% vs 24%**) and technical skills below it (33% vs 39%).

Python · scikit-learn · Pandas · Matplotlib · Seaborn · Machine Learning · Data Analysis

## *Education*

**Bachelor of Engineering in Computer Engineering** — Srinakharinwirot University (SWU) · 2024 – Present · Expected 2027
Relevant coursework: Artificial Intelligence, Healthcare AI Systems, Database Systems, Software Development

Google Cloud, Big Data & GenAI Training — Google Cloud, 2026

## *Leadership & Teaching*

**Computer Engineering Student Leader** — Dept. of Computer Engineering, SWU · 2024 – Present

- Represent CPE students at faculty meetings; plan and run workshops and student-development programs.

**Robotics Teaching Assistant** — Srinakharinwirot University · 2026

- Taught robotics and programming fundamentals to high-school students in hands-on sessions.

## *Skills*

**ML / DL:** PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, OpenCV, ONNX Runtime, Gemini API
**Methods:** Demand forecasting, transfer learning, threshold tuning, time-based validation, Monte Carlo, MLOps
**Data:** Pandas, NumPy, Matplotlib, Seaborn, MySQL, Google Colab
**Languages & web:** Python, SQL, TypeScript, JavaScript, Java, C/C++, FastAPI, React, Next.js, Firebase, Google Cloud, Git